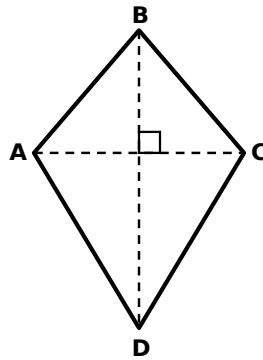


Kite, Trapezium & the Family

Two more quadrilaterals — the kite and the trapezium — and a Venn diagram showing how every quadrilateral type fits together.

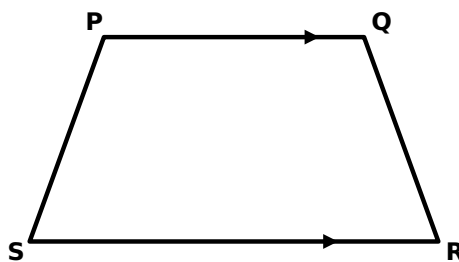
The kite



Kite ABCD: $AB = BC$ and $CD = DA$.

- A kite has **two pairs of adjacent (neighbouring) sides equal**: $AB = BC$ and $CD = DA$.
- One pair of **opposite angles are equal** (the angles between the unequal sides).
- The **diagonals meet at right angles**; the main diagonal **bisects the other** and bisects the two corner angles it passes through.

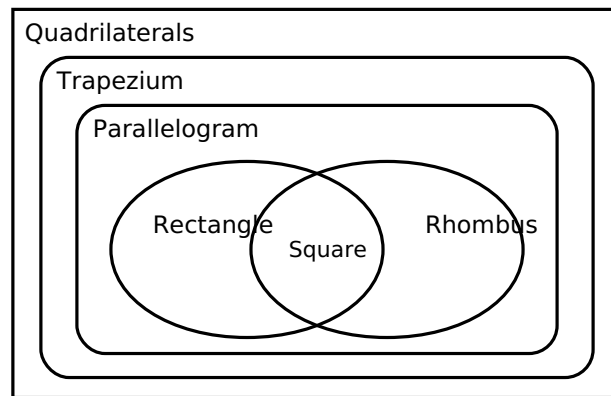
The trapezium



Trapezium PQRS with $PQ \parallel SR$ (shown by arrows).

- A trapezium has **at least one pair of parallel opposite sides**.
- The angles on the same leg (between the parallel sides) are **supplementary**: $\angle P + \angle S = 180^\circ$ and $\angle Q + \angle R = 180^\circ$.
- **Isosceles trapezium**: the non-parallel sides are equal, the base angles are equal, and the diagonals are equal.

The quadrilateral family



Square = rectangle AND rhombus. Every rectangle, rhombus, and square is a parallelogram, and (inclusive view) every parallelogram is a trapezium.

- A **square** sits in the overlap — it is both a rectangle and a rhombus.
- A **kite** is separate, but a kite that is also a parallelogram is a **rhombus**.

Where marks slip away

Trap 1 — Kite $\neq$ rhombus. A kite has two pairs of *adjacent* equal sides; a rhombus has *all four* equal. Not every kite is a rhombus.

Trap 2 — Trapezium angles. The supplementary pairs are the two angles on the *same leg* (between the parallel sides), not any two angles.

Trap 3 — Only one pair equal in a kite. Only **one** pair of opposite angles is equal, not both.

Trap 4 — Isosceles trapezium. "Isosceles" refers to the equal *non-parallel* sides (the legs), which give equal base angles.

Slip 5 — Name the figure precisely (e.g. "isosceles trapezium") to earn full marks.

Model answers — write them like this

Example A. In trapezium PQRS, $PQ \parallel SR$. If $\angle P = 120^\circ$ and $\angle Q = 100^\circ$, find $\angle R$ and $\angle S$.

Step 1: $\angle P$ and $\angle S$ are on the same leg PS, so they are supplementary: $\angle S = 180^\circ - 120^\circ = 60^\circ$.

Step 2: $\angle Q$ and $\angle R$ are on leg QR: $\angle R = 180^\circ - 100^\circ = 80^\circ$.

$$\therefore \angle R = 80^\circ, \angle S = 60^\circ$$

Example B. In kite ABCD ($AB = BC$, $CD = DA$), $\angle B = 100^\circ$ and $\angle D = 40^\circ$. Find $\angle A$ and $\angle C$.

Step 1: All angles add to 360° : $\angle A + \angle B + \angle C + \angle D = 360^\circ$.

Step 2: In this kite $\angle A = \angle C$ (the equal pair), so $\angle A + \angle C = 360^\circ - 100^\circ - 40^\circ = 220^\circ$.

Step 3: $\angle A = \angle C = 220^\circ \div 2 = 110^\circ$.

$$\therefore \angle A = \angle C = 110^\circ$$

Example C. Which quadrilateral is both a rectangle and a rhombus?

Step 1: A rectangle has four right angles; a rhombus has four equal sides.

Step 2: A figure with both is a square.

$$\therefore \text{A square}$$

Next: [try the worksheet](#), then [check the answer key](#).